

SOC Enrichment Pipeline using TheHive & Cortex

Automated Threat Intelligence Enrichment with Real-World Observable Testing

Project Overview

This project demonstrates the **end-to-end integration of Cortex with TheHive** to build a **centralized threat-intelligence enrichment platform** for SOC operations.

Instead of manually switching between multiple external tools (URLScan, GeoIP, MISP, Passive DNS, etc.), this setup enables analysts to:

- Trigger multiple analyzers **from a single Platform TheHive**
- Automatically enrich observables (IP, URL, domain, hash)
- View results directly inside case context in TheHive Platform itself
- Maintain investigation traceability and audit history

The project focuses on **practical SOC workflows**, analyzer configuration, platform integration, and real observable testing based on a simulated attacker IP.

Why Cortex ↔ TheHive Integration?

In real SOC environments:

- Analysts waste time switching tabs between tools
- Context is lost between enrichment steps
- Evidence is scattered across platforms

Cortex + TheHive solves this by:

- Centralizing enrichment inside TheHive cases
- Running analyzers asynchronously via Cortex
- Maintaining results, timestamps, and analyst identity
- Supporting automation, scaling, and future SOAR workflows
- Helping Maintain Compliance & Audit Requirements

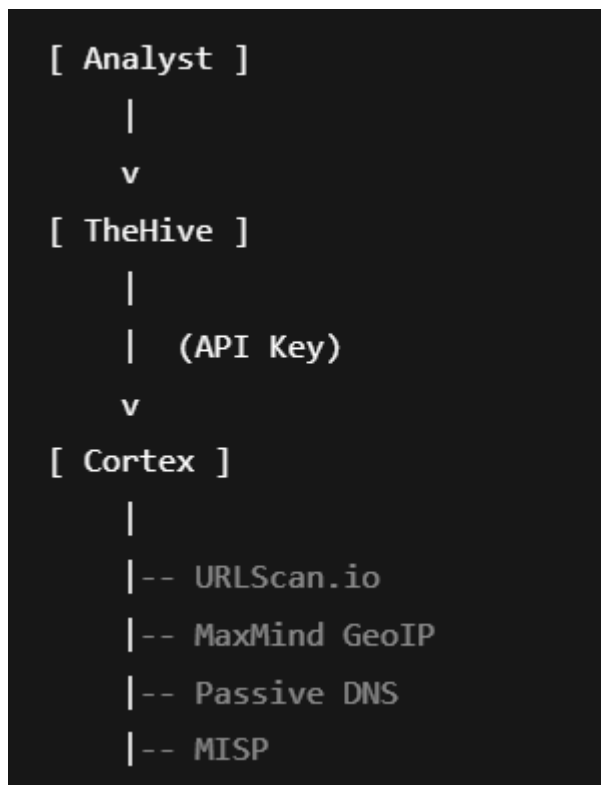
This integration turns TheHive into a **single-pane-of-glass investigation platform**.

Tools & Technologies Used

- **TheHive 5.5.2-1** – Case management & investigation platform
- **Cortex 3.1.8-1** – Analyzer & responder engine
- **Docker & Docker Compose** – Containerized deployment
- **Elasticsearch 7.17.x** – Cortex backend
- **MISP 2.5.x** – Threat intelligence sharing platform

- **urlscan.io** – URL & infrastructure analysis
- **MaxMind GeoIP** – IP geolocation
- **CERT.at Passive DNS** – Passive DNS intelligence

Architecture Overview



- TheHive sends observables to Cortex via API
- Cortex launches analyzer Docker jobs dynamically
- Results are stored and returned to TheHive
- All enrichment appears inside the case timeline inside one single platform helping analysts Effectively

Step-by-Step Implementation

Step 1 – Configuring Analyzers in Cortex

Purpose:

Enable and configure community/free analyzers that do not require paid subscriptions.

Actions Performed:

- Enabled analyzers such as:
 - MaxMind_GeoIP
 - CERTatPassiveDNS
 - Urlscan_io_Search
 - MISP_2_1
- Configured analyzer options:

- TLP & PAP enforcement
- Job timeout
- Observable extraction
- Verified analyzers appear under **Cortex → Analyzers**

Outcome:

Cortex successfully loaded and exposed configured analyzers.

Enable analyzer Urlscan_io_Search_0_1_1

Base details

Name

Urlscan_io_Search_0_1_1

Options

Enable TLP check

True

False

Max TLP

AMBER

▼

Enable PAP check

True

False

Max PAP

AMBER

▼

HTTP Proxy

HTTPS Proxy

CA Certs

Job cache

10

Job timeout

30

Extract observables

True

False

Set to True to enable automatic observables extraction from analysis reports.

Rate Limiting

-- choose unit --

▼

Define the maximum number of requests and the associated unit if applicable.

Cancel

* Required field

Save

Enable analyzer MISP_2_1

Base details

Name

MISP_2_1

Configuration

name

1.

MISP

×

Name of MISP servers

url *

1.

https://

×

URL of MISP servers

key *

1.

×

API key for each server

cert_check *

True

False

Verify server certificate

cert_path

1.

×

Path to the CA on the system used to check server certificate

CERTatPassiveDNS_2_0 Version: 2.0 Author: Nils Kuhnert, CERT-Bund License: AGPL-V3 Type: Docker Checks CERT.at Passive DNS for a given domain.	TLP-AMBER	PAP-AMBER	None	10 Minutes	Edit	Disable
CIRCLHashlookup_1_1 Version: 1.1 Author: Mikael Keri License: AGPL-V3 Type: Docker CIRCL hashlookup uses a public API to lookup hash values against databases of known good files	TLP-AMBER	PAP-AMBER	None	10 Minutes	Edit	Disable

Step 2 – Analyzer Exploration & Selection

Purpose:
Explore available analyzers for learning and understand overlap/redundancy.

- Actions Performed:**
- Reviewed ~250 available analyzers
 - Enabled multiple free/community analyzers for exploration
 - Identified overlapping analyzers (e.g., multiple WHOIS sources)
 - Planned to retain only **high-value analyzers** for long-term SOC usage

Outcome:
Clear understanding of which analyzers are practical vs exploratory.

Cortex

+ New Analysis

Jobs History

Analyzers

Responders

Organization

cortexorg/abducortexocsoar

Analyzers (8)

Data Types (15)

Analyzer

Search

Clear

Page size

50 / page

[AIL_OnionLookup_1_0](#)

Version: 1.0 Author: Fabien Bloume, StrangeBee License: AGPL-V3

Checks the existence of Tor hidden services and retrieving their associated metadata. Onion-lookup relies on an AIL instance to obtain the metadata.

Applies to: domain url fqdn

Run

[Abuse_Finder_3_0](#)

Version: 3.0 Author: CERT-BDF License: AGPL-V3

Find abuse contacts associated with domain names, URLs, IPs and email addresses.

Applies to: ip domain fqdn url mail

Run

[CERTatPassiveDNS_2_0](#)

Version: 2.0 Author: Nils Kuhnert, CERT-Bund License: AGPL-V3

Checks CERT.at Passive DNS for a given domain.

Applies to: domain fqdn ip

Run

[CIRCLHashlookup_1_1](#)

Version: 1.1 Author: Mikael Keri License: AGPL-V3

CIRCL hashlookup uses a public API to lookup hash values against databases of known good files

Applies to: hash

Run

[ClamAV_FileInfo_1_1](#)

Version: 1.1 Author: Brian Laskowski License: AGPL-V3

Use Clamscan with custom rules

Applies to: file

Run

[MISP_2_1](#)

Version: 2.1 Author: Nils Kuhnert, CERT-Bund License: AGPL-V3

Query multiple MISP instances for events containing an observable.

Applies to: domain ip url fqdn url_path user-agent hash mail mail_subject registry regexp other filename mail-subject

Run

[MaxMind_GeolIP_4_0](#)

Version: 4.0 Author: CERT-BDF License: AGPL-V3

Use MaxMind to geolocate an IP address.

Applies to: ip

Run

[Urlscan_io_Search_0_1_1](#)

Version: 0.1.1 Author: ninoseki, Kyle Parrish (@armydo) License: MIT

Search IPs, domains, hashes or URLs on urlscan.io

Applies to: ip domain hash fqdn url

Run

50 / page

TheHive Project 2016-2021, AGPL-V3

Version: 3.1.8-1

Step 3 – Integrating Cortex with TheHive

Purpose:

Allow TheHive to trigger Cortex analyzers directly from cases.

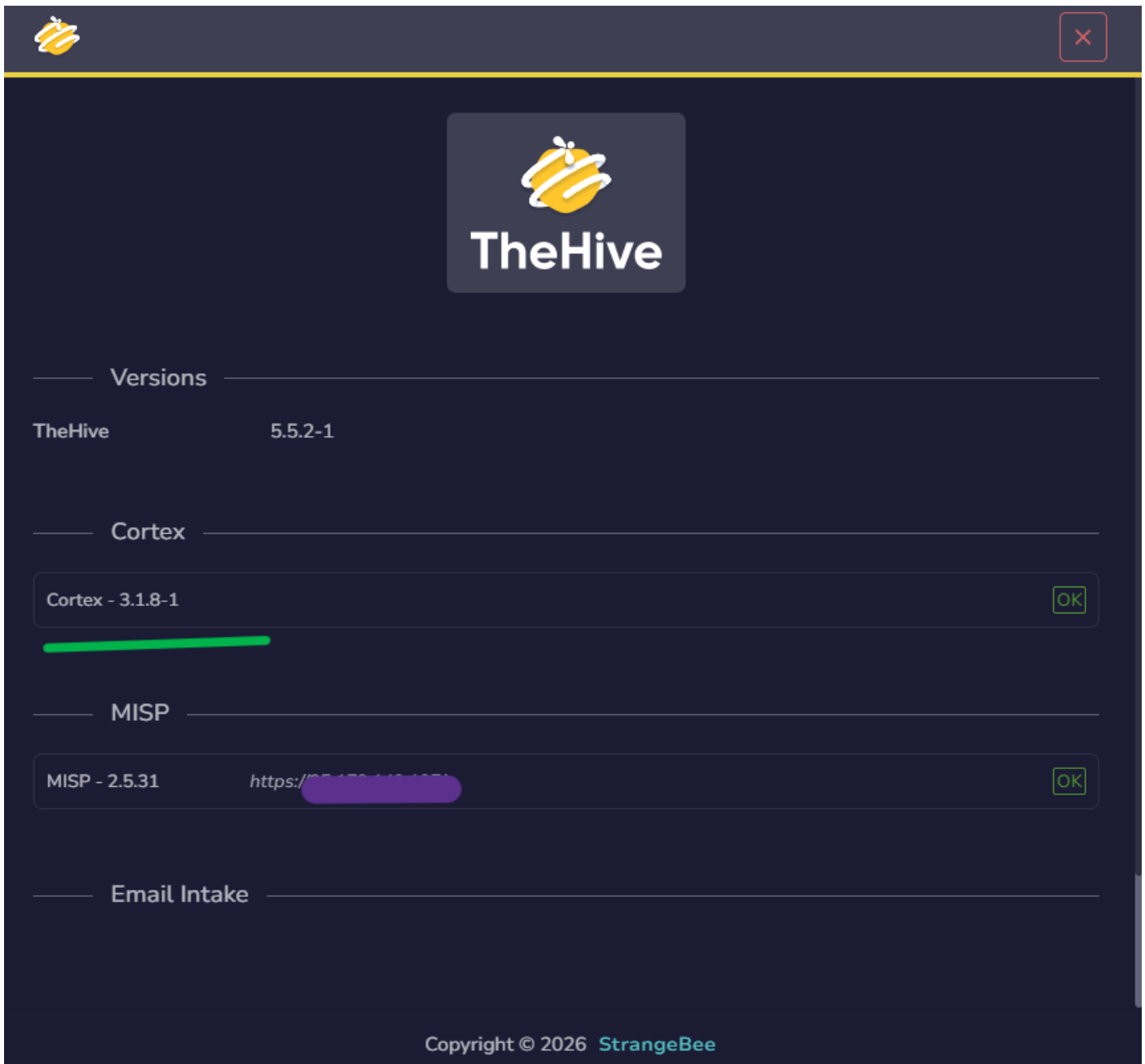
Actions Performed (via TheHive UI):

- Navigated to **Platform Management** → **Connectors** → **Cortex**
- Added Cortex server details:
 - Server URL
 - Cortex API Key
- Configured SSL and timeout options
- Successfully saved and validated connection

Outcome:

TheHive successfully connected to Cortex (status: OK).

The screenshot displays the 'Platform Management / Connectors / Cortex' configuration page in TheHive. The interface is divided into two main sections: a left sidebar with navigation options (License, Status, Branding, Connectors, Authentication, SMTP, Global Endpoints, LDAP servers) and a main content area. The 'Connectors' section is active, showing a list of connectors with 'Cortex' selected. Below the list, there are input fields for 'Server name' (containing '3'), 'Refresh Delay' (5 seconds), 'Frequency of status checks' (1 minute), and 'Maximum time TheHive waits for the job to complete' (3 hours). A 'Servers' section at the bottom shows a table with one entry for 'Cortex'. A success message box in the bottom left corner states: 'success Cortex configuration has been successfully saved'. On the right side, a detailed configuration panel for the 'Cortex' connector is visible, showing fields for 'Server name', 'Server url' (http://), and 'API Key' (masked with dots). Below these are 'Proxy settings' (Use default configuration, Enabled, Disabled) and 'SSL Settings' (Do not check Certificate Authority, Recommended, Disable hostname Verification).



Step 4 – Verifying Analyzer Visibility in TheHive

Purpose:

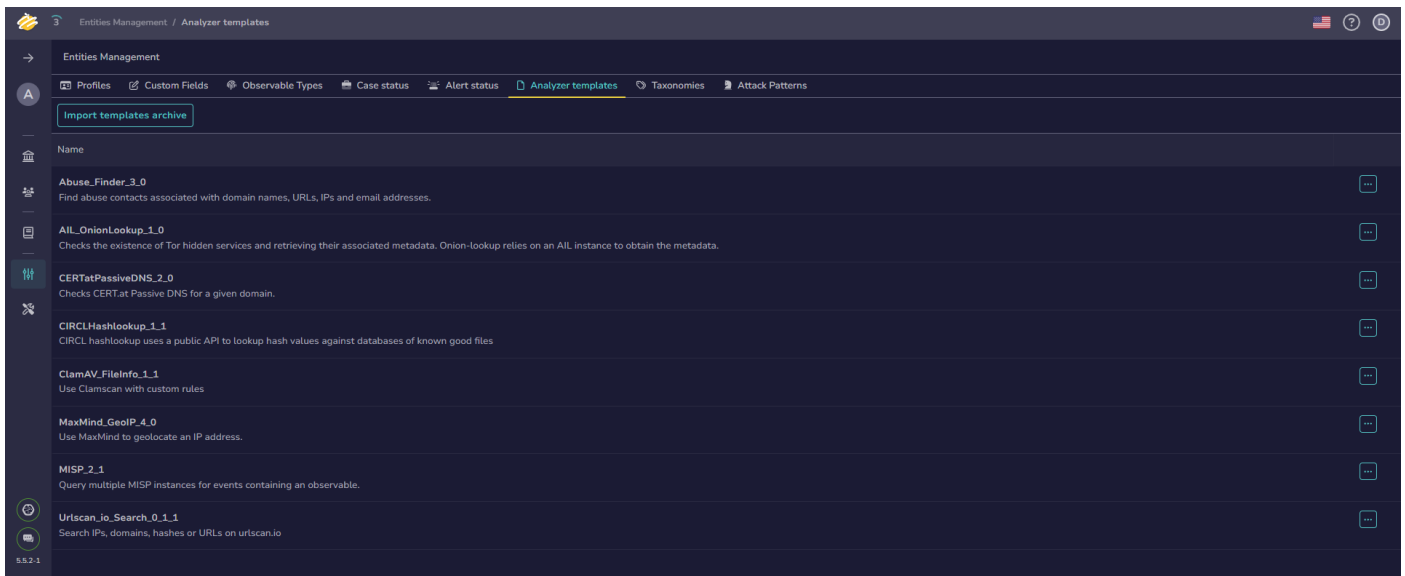
Ensure analyzers enabled in Cortex are visible inside TheHive.

Actions Performed:

- Opened **Entities Management** → **Analyzer Templates**
- Verified Cortex analyzers were listed and selectable

Outcome:

All configured Cortex analyzers appeared in TheHive.



Step 5 – Case Creation & Observable Testing

Purpose:

Test the full pipeline using a realistic SOC scenario.

Scenario:

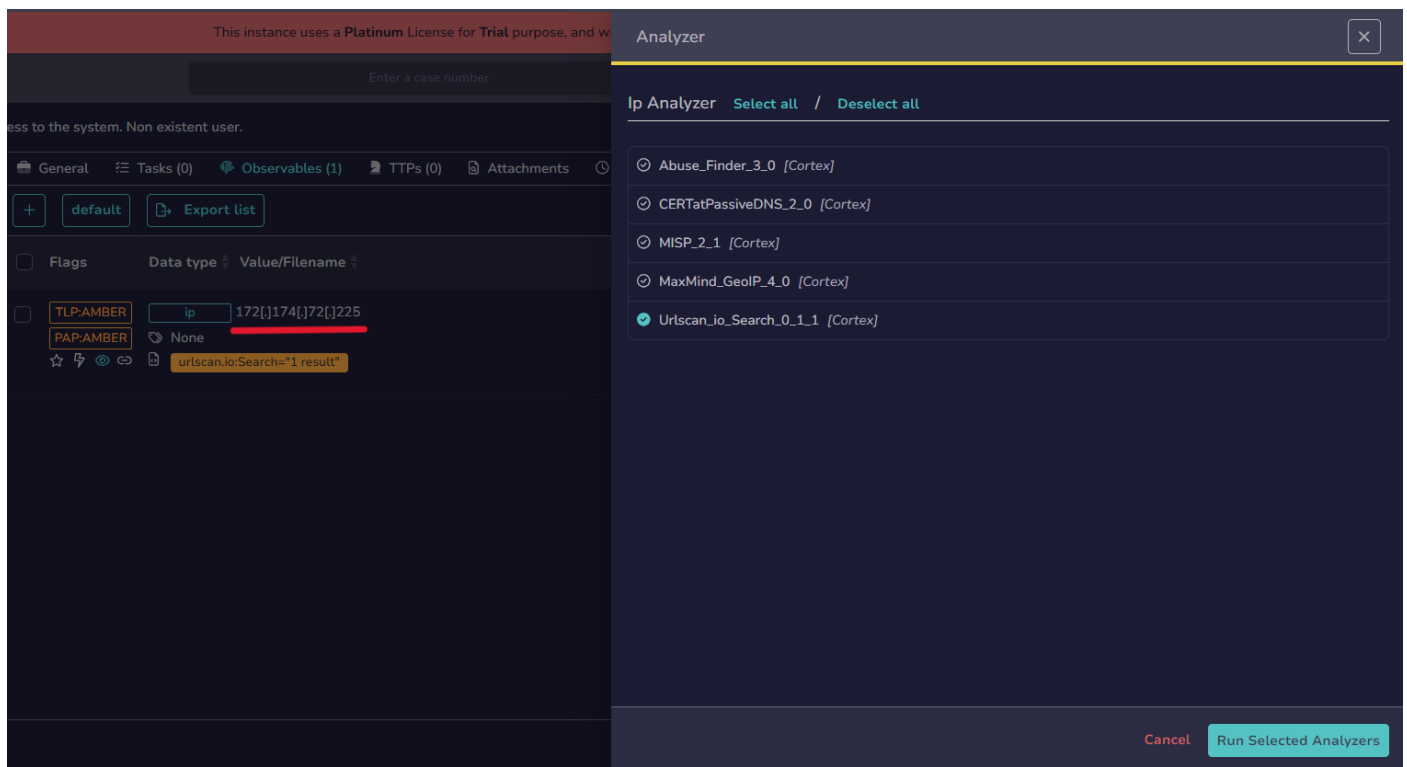
- Attacker IP observed attempting SSH brute-force activity
- IP added as an **observable** in a TheHive case

Actions Performed:

- Created a new case
- Added attacker IP as observable
- Selected multiple analyzers:
 - Urlscan_io_Search
 - MaxMind_GeoIP
 - MISP
 - CERTatPassiveDNS

Outcome:

Analyzers successfully triggered from TheHive.



Step 6 – Analyzer Results & Raw Output Analysis

Purpose:

Validate analyzer execution and understand returned intelligence.

Observations:

- Urlscan.io returned:
 - Related domains
 - Infrastructure details
 - Country and hosting provider
 - Historical scan data
- Results visible in:
 - Structured report
 - Raw JSON output

Outcome:

Rich contextual intelligence obtained directly inside the case.

Analysis report

Hide raw result

```
{
  "type": "ip",
  "query": "172.174.72.225",
  "service": "get",
  "indicator": {
    "results": [
      {
        "task": {
          "visibility": "public",
          "method": "automatic",
          "domain": "filesharee.ddns.net",
          "apexDomain": "ddns.net",
          "time": "2022-11-19T13:23:19.522Z",
          "source": "openphish",
          "uuid": "a486d3a0-dc00-447c-acc8-fc6ed4123e25",
          "url": "http://filesharee.ddns.net/"
        },
        "stats": {
          "uniqIPs": 8,
          "uniqCountries": 3,
          "dataLength": 1025736,
          "encodedDataLength": 682125,
          "requests": 20
        },
        "page": {
          "country": "US",
          "server": "Apache",
          "ip": "172.174.72.225",
          "domain": "filesharee.ddns.net",
```

Analysis report



Show raw result

urlscan.io search results

URL	Result	Time
http://filesharee.ddns.net/	https://urlscan.io/result/a486d3a0-dc00-447c-acc8-fc6ed4123e25	2022-11-19T13:23:19.522Z

Step 7 – External Intelligence Validation

Purpose:

Cross-verify analyzer output manually for learning and confidence.

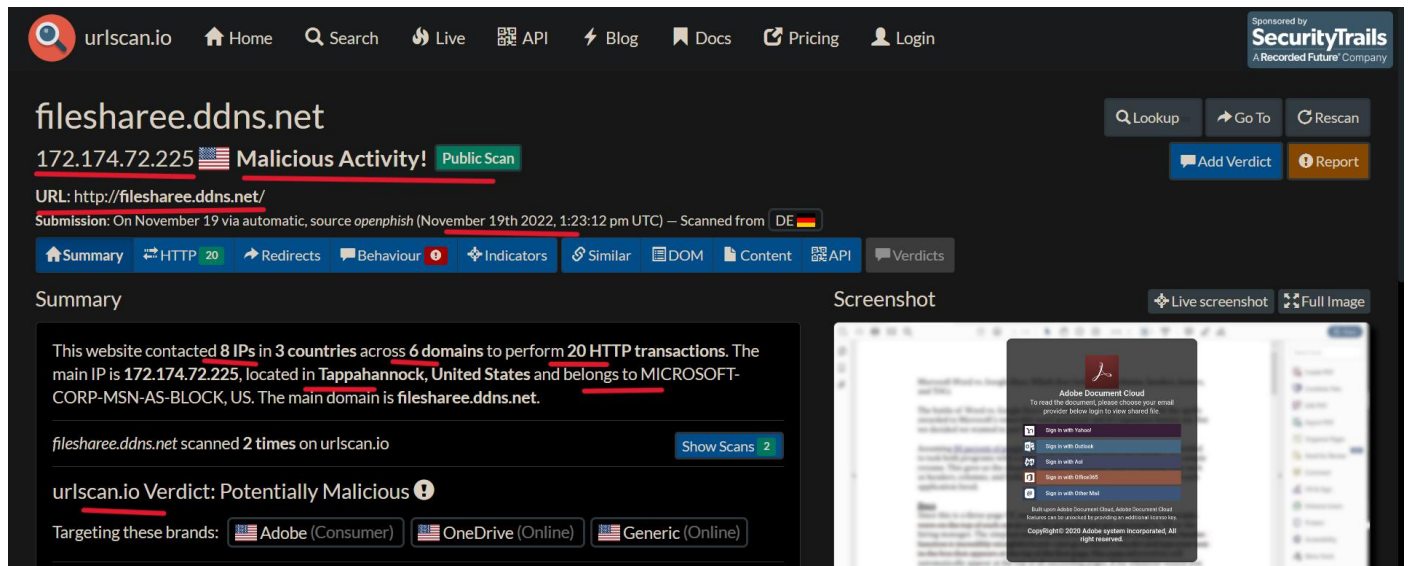
Actions Performed:

- Followed urlscan.io link provided in Cortex result

- Verified:
 - Malicious verdict
 - Infrastructure mapping
 - Domain relationships
- Confirmed Cortex data matched external source

Outcome:

Analyzer output was accurate and reliable.



Step 8 – Job Tracking & Audit Validation

Purpose:

Ensure traceability across platforms.

Actions Performed:

- Checked **Cortex → Job History**
- Verified:
 - Job status
 - Observable value
 - Timestamp
 - Triggering user
- Confirmed same job visibility inside TheHive

Outcome:

Full audit trail maintained across Cortex and TheHive.

Cortex + New Analysis Jobs History Analyzers Responders Organization cortexorg/abdulcortexsocsoar

Jobs History (1)

Data Types (15) Select Job Type (2) Select Analyzers (8) Select Observable Search for observable data Search Clear 50 / page

Status	Job details	TLP	PAP
Success	[ip] 172[.]174[.]72[.]225 Analyzer: Urlscan_io_Search_0_1_1 Date: 4 minutes ago User: cortexorg/underscore@underscore.co	TLP-AMBER	PAP-AMBER

50 / page

Cortex + New Analysis Jobs History Analyzers Responders Organization cortexorg/abdulcortexsocsoar

Job details

Urlscan_io_Search_0_1_1

Artifact
[IP] 172[.]174[.]72[.]225

Date
an hour ago

TLP
TLP-AMBER

PAP
PAP-AMBER

Status
Success

Report summary
urlscan.io:Search=1 result

Job report

Parameters

```
{
  "organisation": "abdulethehiveorg",
  "user": "opensource@opensource.co"
}
```

Report

```
{
  "summary": {
    "taxonomies": [
      {
        "level": "suspicious",
        "namespace": "urlscan.io",
        "predicate": "Search",
        "value": "1 result"
      }
    ]
  },
  "full": {
    "type": "ip",
    "query": "172.174.72.225",
    "service": "get",
    "indicator": {
      "results": [
        {
          "task": {
            "visibility": "public",
            "method": "automatic",

```

Step 9 – Multi-Analyzer Correlation (Learning Phase)

Purpose:

To understand how multiple analyzers enrich the same observable and how TheHive consolidates intelligence received from Cortex.

Actions Performed:

- Ran additional analyzers on the same attacker IP for learning purposes
- Executed analyzers including: Abuse_Finder, CERTatPassiveDNS, MaxMind_GeoIP, MISP, and Urlscan.io
- Reviewed how each analyzer provided different intelligence perspectives such as geolocation, passive DNS, threat intelligence matches, and infrastructure behavior

Observation:

- Not all analyzers return results for every observable, which reflects real SOC behavior
- Some analyzers failed due to external service limitations, rate limits, or data unavailability
- TheHive clearly displayed successful and failed analyzers with timestamps, maintaining analyst visibility and confidence

Outcome:

- Gained clarity on analyzer redundancy versus unique value

- Identified high-signal analyzers suitable for production use
- Reinforced the importance of selective analyzer enablement in SOC workflows

Reports		
MaxMind:Location="Germany/Europe... urlscan.io:Search="1 result" MISP:Search="1 event(s)" Abuse_Finder:Address="abuse@micr...		
Description		
Not Specified		
Analyzers		
Analyzer	Last analysis	
Abuse_Finder_3_0	✓ 30/01/2026 21:52	
CERTatPassiveDNS_2_0	⚠ 30/01/2026 21:53	
MaxMind_GeolP_4_0	✓ 30/01/2026 21:52	
 MISP_2_1	✓ 30/01/2026 22:30	
Urlscan_io_Search_0_1_1	✓ 30/01/2026 20:40	

Project Outcome

- ✓ Cortex analyzers successfully configured and executed
- ✓ Seamless TheHive ↔ Cortex integration
- ✓ Real-world observable enriched using multiple intelligence sources
- ✓ Results centralized within a single SOC workflow
- ✓ Demonstrated SOC-ready investigation pipeline

Skills & Concepts Gained

- SOC enrichment architecture design
- Cortex analyzer lifecycle & execution model
- TheHive case-based investigation workflow
- Threat intelligence correlation
- Docker-based SOC tooling
- Practical SOAR foundation knowledge

❏ Conclusion

This project demonstrates how **TheHive and Cortex together form a powerful SOC enrichment platform**, eliminating manual effort and improving investigation efficiency.

By integrating multiple intelligence sources into a single workflow, analysts gain:

- Faster triage
- Better context
- Stronger decision-making
- Audit-ready investigations

This setup closely mirrors **real-world SOC environments** and serves as a strong foundation for future automation, response playbooks, and SOAR expansion.
